

This assignment will not be graded and is only for practice.

Level 1

1) Choose the correct options.

1 point

- ☐ Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- ☐ Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- ☐ Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.
- ☐ Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $\langle a, b \rangle = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

2) If A is the matrix representation of an orthogonal transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then choose the set of correct statements. **1 point**

- ☐ $AA^T = I$
- ☐ $A = A^T$
- ☐ AA^T is an upper triangular matrix
- ☐ AA^T is a lower triangular matrix
- ☐ $A^T = A^{-1}$
- ☐ $A^T = -A^{-1}$

- ☐ A^{-1} does not exist
- ☐ A^{-1} is also an orthogonal matrix

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$AA^T = I$$

AA^T is an upper triangular matrix

AA^T is a lower triangular matrix

$$A^T = A^{-1}$$

A^{-1} is also an orthogonal matrix

3) If A is the matrix representation of an orthogonal transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, **1 point**
then choose the set of correct statements.

- ☐ The column vectors of A are orthogonal but not orthonormal.
- ☐ The column vectors of A are orthonormal.
- ☐ The row vectors of A are orthogonal but not orthonormal.
- ☐ The row vectors of A are orthonormal.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The column vectors of A are orthonormal.

The row vectors of A are orthonormal.

Level 2

4) Consider the orthogonal set $B = \{(1, 1, 1), (-1, 1, 0)\}$ in $\mathbb{R}^3$. Find a third vector in $\mathbb{R}^3$ which is orthogonal to both the vectors in B . **1 point**

- ☐ (1, 1, -2)
- ☐ (1, -1, 2)
- ☐ (-1, 1, -2)
- ☐ (-1, -1, -2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1, 1, -2)

5) Consider the orthogonal set $B = \{(1, 1, 1), (-1, 1, 0)\}$ and vector obtained in question 4 which is orthogonal to the set B . Form the columns of an orthogonal matrix U by scaling these vectors appropriately. Choose the correct option. **1 point**

- ☐ $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$
- ☐ $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$
- ☐ $U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$
- ☐ $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$$

6) Consider a vector $v = (3, 0, 4)$ and U from Question 5. Choose the correct option **1 point** using the dot product as the standard inner product.

- ☐ $\|v\| = \|Uv\|$
- ☐ $\|v\| = \|Uv\|$
- ☐ $2\|v\| = \|Uv\|$
- ☐ $\|v\| = 2\|Uv\|$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\|v\| = \|Uv\|$$

7) Let the matrix representation of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be $U =$ **1 point**

$$\begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}.$$

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

- ☐ T is an orthogonal transformation.
- ☐ T is not an orthogonal transformation.
- ☐ B is a real symmetric matrix.
- ☐ $U^{-1}BU$ is a diagonal matrix.
- ☐ $U^{-1}BU$ is a scalar matrix.

- ☐ $U^{-1}BU$ is an upper triangular matrix.
- ☐ $U^{-1}BU$ is a lower triangular matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is an orthogonal transformation.

B is a real symmetric matrix.

$U^{-1}BU$ is a diagonal matrix.

$U^{-1}BU$ is an upper triangular matrix.

$U^{-1}BU$ is a lower triangular matrix.

Your score is: 0/7